L'ORÉAL SUSTAINABILITY CHALLENGE 2026 · RESEARCH & INNOVATION (OPTION 02)

SKINNOVA · Climate-Adaptive Skin Longevity Platform for Indian GenZ

BRAND PORTFOLIO RELEVANCE

L'Oréal Paris · CeraVe · Garnier

Zero-Backend PWA · On-Device Image Analysis · Live Open-Meteo

01 · SCAN

On-Device Facial Analysis

Forehead & cheek patches in CIELAB. Visible-signal proxies only. Image remains on user device.

02 · CLIMATE

Open-Meteo Telemetry

Live ambient UV, PM2.5, temp & humidity. 8 Indian city presets with sample-data fallback.

03 · FUSION

CSSI & Guidance Index

Transparent environmental index fused with baseline-relative visible signal comparisons.

04 · ROUTINE

Minimal Adaptive Routine

AM/PM barrier essentials (SPF, Niacinamide, Ceramides). 'Skip-Today' on low-stress days.

VISUAL PIPELINE: Environmental Exposure ➔ Visible Skin Signals ➔ Personalised Guidance ➔ Minimal Adaptive Routine ➔ Lower Overconsumption

THE CLIMATE CHALLENGE FOR INDIAN GENZ

Three Key Environmental Stressors

- **UV Exposure:** Intense solar irradiance inducing oxidative stress pathways and photo-induced pigmentation concerns in melanin-rich skin.
- **Air Pollution:** Ambient particulate matter (PM2.5/PM10) exposure accelerating surface sebum oxidation and cellular barrier fatigue.
- **Heat & Humidity:** Monsoon humidity (>80% RH) and tropical heat increasing sebum flux, follicular discomfort, and barrier challenges under multi-product layering.

SCIENTIFIC ENGINE · DECISION SUPPORT

Transparent Environmental-to-Routine Framework

```
CSSI = 0.35 × S(UV) + 0.35 × S(PM2.5) + 0.20 × S(RH_Heat) + 0.10 × S(Temp)
```

Proposed prototype weights; not clinically validated biological weights.

Experimental Personalised Skin Guidance Index:

```
Guidance Index = 100 − (0.55 × CSSI + 0.45 × Visible Signal Index)
```

Experimental recommendation heuristic; not a clinical measure of skin health or biological aging.

- **Future Clinical Validation:** Protocol planned against Corneometer® (hydration), Tewameter® (TEWL), and Mexameter® (erythema) instruments.

SUSTAINABILITY · RESPONSIBLE INNOVATION

Circular Design & Anti-Overconsumption

- **Circular Packaging Concepts:** Refillable and recyclable packaging designs with modular replaceable inner cartridges to minimize single-use plastics.
- **Responsible Botanical Sourcing:** Sourcing framework structured around applicable Indian Biological Diversity Act Access and Benefit Sharing (ABS) considerations.
- **Minimalist Routine Support:** 'Skip-Today' recommendations on low-stress days to actively discourage unnecessary product layering.
- **Privacy-First On-Device Compute:** Local processing avoids continuous cloud GPU infrastructure energy demands.

CORE DIFFERENTIATOR · ETHICAL ON-DEVICE AI

Fairness & Privacy Architecture

- **Baseline-Relative Analysis:** Designed to reduce absolute skin-tone dependence through individual baseline-relative comparisons (ΔL*, Δa*, ΔLaplacian).
- **On-Device Processing:** All canvas computer vision executes locally on the user's phone; zero facial images uploaded or stored on cloud servers.
- **Lighting-Quality Checks:** Automated checks for underexposure, glare, and cheek asymmetry prompt friendly retakes rather than issuing misleading scores.

PAIRED FORMULATION CONCEPT

Lightweight Antioxidant + Barrier-Support Gel-Cream

Target Sensory: Lightweight, humidity-friendly, non-comedogenic formulation target.

- **Proposed Ingredient Classes:** Niacinamide, Ceramides, Vitamin C/E derivatives, and ethically sourced Indian botanicals such as amla (Phyllanthus emblica).
- **Concept Notice:** Concept formulation — not an existing or clinically validated product.

GO-TO-MARKET · BRAND PORTFOLIO ALIGNMENT

Scalable Engagement for Indian GenZ

- **Target Demographic:** Indian GenZ, initially urban college students and young professionals seeking clinical clarity and climate-adapted barrier defense.
- **Frictionless Digital Entry:** Designed for fast QR-based mobile web access; requires zero app store download or mandatory user login.
- **Activation Touchpoints:** Interactive demonstrations at university campuses and beauty retail settings (smart mirror kiosks in partner stores).
- **Brand Portfolio Synergy:** Potential conceptual fit for CeraVe (barrier care), L'Oréal Paris (youth longevity), or Garnier (botanical sourcing).

Concept proposal; metrics and formulation targets pending validation. Not a medical diagnostic tool. | **Explore Interactive Prototype:** https://koushik1914.github.io/skinnova/ | **Koushik Voota | Amrita Vishwa Vidyapeetham**